MD MEHEDI HASAN

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Objective

My master's thesis — defended July 14, 2026, at Saga University — developed and validated a two-stage RFID-based Digital Twin framework for emergency department resource management in Bangladesh hospitals. All three research questions were answered and supported by real data from 27 simulation log files: 99.7% alert resolution (RQ3), Adaptive Authority Engine converging nurse staffing within 4–5 simulated days (RQ2), and Unity 3D demonstrating superior situational awareness across 5 dimensions (RQ1). At the PhD level, I aim to extend this toward adaptive, real-time intelligent systems integrating machine learning with RFID localization and Digital Twin frameworks for resource-constrained healthcare environments. I am seeking a fully funded PhD position in healthcare informatics, biomedical engineering, or related fields.

Education

Saga University *Oct 2024 – Sep 2026 Saga, Japan*

Master of Engineering — Computer Science & Information Technology

Supervisor: Professor Eisuke Hanada · GPA: 3.9 / 4.0 (Final semester: 4.0 / 4.0)

English proficiency: TOEFL scheduled August 2026 (target 85+)

Thesis (defended July 14, 2026): A Digital Twin Approach to Emergency Department Resource Management

A two-stage Digital Twin framework combining a MATLAB agent-based simulation (C1+C2) with a Unity 3D VR visualization layer (C3) for RFID-based localization and resource tracking across a 20×15 m ward. All three RQs answered from 27 simulation log files: 100% zone-assignment accuracy (RQ3), Adaptive Authority Engine converging staffing within 4–5 days (RQ2), Unity 3D demonstrating superior situational awareness across 5 dimensions (RQ1)

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Bangladesh Army International University of Science & Technology (BAIUST) *2018 – 2022 Cumilla, Bangladesh*

Bachelor of Science — Computer Science & Engineering

GPA: 3.16 / 4.0

Supervisors: Mohammad Asaduzzaman Khan (Assoc. Prof. & Head, CSE); Sabina Zaman (Lecturer, CSE)

Research Experience

Hanada Laboratory, Saga University *Oct 2024 – Sep 2026 Saga, Japan*

Graduate Researcher

- C1 — MATLAB Agent-Based Simulation: Operational RFID-based ED simulator with 3-class FSM agents, stochastic surge generation, nearest-nurse Euclidean dispatch, and CSV logging; validated across 19 MATLAB runs (39–762 patient arrivals), 99.7% alert resolution, 83.0s mean response time.
- C2 — Adaptive Authority Engine: KPI-driven bang-bang staffing controller converging to a [20, 40] min wait-time deadband within 4–5 simulated days; zero incorrect decisions across 16 simulated days; full audit trail for clinical governance in resource-constrained settings.
- C3 — Unity 3D VR Transformation: 11 C# scripts, NavMesh pathfinding, RFID pulse animation, 3 role-based views (Admin / Nurse-Level / Equipment Manager); 100% alert response coverage and 2.7× surge modulation accuracy vs. MATLAB 2D baseline.
- C4 — Integrated MATLAB→Unity Framework: Reproducible, free-software Digital Twin pipeline; directly applicable to Bangladeshi and LMIC hospital planning contexts without hardware investment. All three RQs answered from 27 simulation log files.
- Supported by Research Assistantship (RA) from the laboratory research grant throughout the MEng programme (¥19,800/month).

International Research Presentation — Saga University × Kasetsart University *Feb 23–27, 2026 Bangkok, Thailand*
Cross-Cultural Academic Collaboration Programme
◦ Presented RFID-based ED simulation research to an international academic audience as part of a bilateral collaboration programme between Saga University (Japan) and Kasetsart University (Thailand).
◦ Participated in cross-cultural academic exchange covering research methodologies and healthcare informatics applications across Southeast Asian contexts.

Patents and Publications

C=Conference · J=Journal · S=In Submission

[C.1] Improving Patient Care in the Emergency Wards of Bangladesh Through an RFID-Based Web Simulation
Hasan MD Mehedi, Hossain Shakil, Eisuke Hanada
20th EAI Int'l Conference on Body Area Networks (EAI BodyNets 2025), Dec 2025 (2025) **[Accepted / In Press]**

[C.2] Digital Twin-Based Emergency Resource Localization System for Crowded Hospital Wards Using RFID and VR Visualization
Hasan MD Mehedi, Hossain Shakil, Eisuke Hanada
IEICE General Conference 2026 — Session B-19, Kyushu Sangyo University, Mar 2026 (2026) **[Published]**

[C.3] Real-Time RFID Tracking Simulation for Hospital Resource Management: A 3D Unity Implementation with Dynamic Surge Response and VR Readiness
Hasan MD Mehedi, Eisuke Hanada
ISMICT 2026 — 20th Int'l Symposium on Medical Information and Communication Technology (2026) **[Under Review]**

[C.4] Cross-Census Stability and Surge Resilience of RFID-IoT Dispatch in a Bangladeshi Emergency Department Simulation
Hasan MD Mehedi, Eisuke Hanada
EAI HealthyAIoT 2026 — 11th EAI Int'l Conference on AIoT Healthcare Systems, Guangzhou, China (2026) **[Under Review]**

[S.1] An RFID-Based Dynamic Resource Management Simulation for Smart Hospital Wards: A Bangladesh Healthcare Perspective
Hasan MD Mehedi, Eisuke Hanada
Journal (Extended from EAI BodyNets 2025 — target submission July 2026) (2026) **[In Preparation]**

Co-authored contributions:

[C.5] Transfer Learning-Based Renal Ultrasound Classification for CAKUT Screening: A Preliminary Study Using ResNet-50
Ahmed Alif, Hasan MD Mehedi, Eisuke Hanada
ISMICT 2026 — 20th Int'l Symposium on Medical Information and Communication Technology (2026) **[Under Review]**

[C.6] A Lightweight AIoT Framework for CAKUT Screening in Resource-Limited Community Clinics: System Design and Edge Deployment Feasibility
Ahmed Alif, Hasan MD Mehedi, Eisuke Hanada
EAI HealthyAIoT 2026 — 11th EAI Int'l Conference on AIoT Healthcare Systems, Guangzhou, China (2026) **[Under Review]**

[C.7] Traffic-Aware Routing and Real-Time Clinical Synchronisation for Mobile Emergency Medical Services in Resource-Constrained Environments
Das Bishal, Hasan MD Mehedi, Eisuke Hanada
ISMICT 2026 — 20th Int'l Symposium on Medical Information and Communication Technology (2026) **[Under Review]**

Teaching, Mentoring & Outreach

Graduate Peer Mentor, Saga University *Apr – Sep 2026 Saga, Japan*
Academic & Settlement Support for International Student — Faculty-Remunerated
◦ Comprehensive peer mentoring covering CS/engineering academic support and practical settlement guidance — paperwork, residence registration, banking, health insurance; remunerated hourly (avg. ¥18,000/month).

Team Leader & English Presentation Coach *May–Jul 2025 & 2026 Saga, Japan*
Saga University × Chienkan High School Science English Outreach Programme
◦ Led two consecutive teams of Japanese high school students from project concept to final scientific presentation across two years.
◦ 2025: scientific poster coaching. 2026: oral presentation restructuring and spoken English delivery coaching.

Projects

Hospital Management System — Mainamati Cantonment General Hospital (MCGH) *Jan – May 2022 Bangladesh*

BSc Final Year Project · BAIUST (Course: CSE-400)

Tools: HTML, CSS, JavaScript, MySQL

- Developed a web-based HMS for a real military hospital in Cumilla, Bangladesh as part of a 5-member team; contributed as frontend developer.
- System covered patient registration, appointment scheduling, ward allocation, and medical records — replacing manual paper-based workflows.
- *Team: Thasin Mahmud, Md. Imran Hossain Emu, Md. Mahin Hossain Munna, Ahamed Jamil Bhuiyan, Md. Mehedi Hasan.*

Skills

Proficient

Simulation & Modeling	MATLAB (Agent-Based, Monte Carlo, V5.3), Unity 3D, Digital Twin architecture
RFID & IoT	Passive UHF RFID, Active RFID, Chipless RFID, RSSI trilateration, LANDMARC
3D Visualization & VR	Unity 3D, NavMesh pathfinding, RFID pulse animation, role-based views
Programming	MATLAB scripting, C# (Unity), Python
Research & Writing	Academic writing, simulation design, literature review, conference presentation, LaTeX

Familiar

Web & Frontend	HTML, CSS, JavaScript
Tools & Platforms	Git / GitHub, Arduino, MS Office
Design Software	Adobe Photoshop, Adobe Illustrator, Adobe After Effects
Data	CSV pipeline, edge-cloud architecture, statistical analysis

Languages

Language	*Listening*	*Reading*	*Speaking*	*Writing*	*Notes*
Bengali	C2	C2	C2	C2	Native
English	B2	C1	B2	B2	Professional — TOEFL scheduled Aug 2026 (target 85+)
Japanese	A2	A2	A2	A1	Daily conversation (approx. JLPT N4; exam not yet taken)
Hindi/Urdu	B1	A1	B1	A1	Conversational; limited reading/writing

Levels: A1/A2 — Basic | B1/B2 — Independent | C1/C2 — Proficient (CEFR) | JLPT: N5 (beginner) → N1 (advanced)

Honors, Scholarships & Awards

MEXT Honors Scholarship — Japanese Government *Oct 2024 – Mar 2025*

Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan

¥48,000/month · Competitive national scholarship

Saga City Scholarship *May 2025 – Mar 2026*

Saga City Government, Japan · ¥20,000/month · Competitive municipal scholarship

Full Tuition Exemption — Saga University *Oct 2024 – Sep 2026*

National university tuition fully waived (~¥535,800/year)

Candidate for Academic Incentive Award *Mar 2026*
IEICE General Conference 2026, Session B-19-01, Kyushu Sangyo University
Paper: Digital Twin-Based Emergency Resource Localization System for Crowded Hospital Wards Using RFID and VR Visualization

Perfect Semester GPA — 4.0 / 4.0 *2025*
Final completed semester (2025 Fall), Saga University

REFERENCES

1. **Professor Eisuke Hanada**
   Faculty of Science and Engineering, Saga University, Japan
   Email: hanada@cc.saga-u.ac.jp · ORCID: 0000-0002-6718-0101
   Relationship: Master's Thesis Supervisor

2. **Professor Yoshiaki Hori**
   Computer and Network Center, Saga University, Japan
   Research: Network Security, Computer System Security, Network Architecture
   Email: hori@cc.saga-u.ac.jp
   Relationship: Second Academic Referee (Master's thesis)

Additional referees available upon request.